

A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Importance & features of energy resources - Set 1

Mark scheme

AQA spec 3.3.1-3.3.2; Energy resources; textbook pp. 170-182

Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	• In general, higher GDP per capita is associated with higher energy use per capita (positive correlation) • But not proportional: B has a high GDP but much lower energy use than A or E; C uses relatively high energy for its GDP	2
01.2	• Affluent people buy more consumer goods, appliances and travel more / drive larger, less efficient vehicles and heat larger homes • Manufacture and transport of the extra products bought uses energy (indirect use); larger houses use more energy to build • Affluent countries can afford energy for non-essential uses; cheap local energy (eg petrol in the USA) encourages wasteful use	3
01.3	<i>Any 2 from:</i> • Higher energy prices / taxes encouraging conservation • Culture of energy conservation (eg Scandinavia after past shortages) / environmental awareness • Milder climate needing less heating or air conditioning • Service-based economy: heavy industry (steel, aluminium) has moved to other countries so energy embodied in imports is not counted • Smaller / more efficient vehicles, better public transport, better insulated buildings	2
01.4	• Decline of heavy industries such as steel and aluminium smelting in the UK • The energy-intensive goods are now manufactured abroad (eg China) so the energy is counted in the exporting country; also efficiency improvements	2
01.5	• Water: pumping, treatment / sterilisation, sewage treatment; Agriculture: machinery, fertiliser manufacture (Haber process), irrigation pumps, processing / refrigeration	1

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 170-174; AQA spec 3.3.1.	
02.1	- The amount of energy contained in a given mass of the resource (or harnessed per kg of equipment for renewables) - High energy density means smaller quantities are needed so storage and transport are easier and cheaper (eg 75 t of aviation fuel carries a 400 t aircraft 5600 km) - High energy density fuels reach high temperatures needed for smelting / steam turbines	3
02.2	$500\,000 / 28 = 17\,857\text{ kg}$ $= 17.9\text{ tonnes}$ (accept 18; book quotes 13 t for 0.7% enriched fuel)	2
02.3	- The electricity they generate can power high-energy-requirement technologies - Electricity can be converted to high energy density secondary fuels, eg hydrogen by electrolysis, or stored (batteries, pumped storage)	2
02.4	- Intermittent: not available all the time, eg tidal (but the times are predictable so alternatives can be planned) - Unpredictable: intermittent and the timing / amount cannot be forecast accurately, eg wind, solar	2
02.5	Gaseous natural gas has a very low energy density by volume; LNG at $-160\text{ }^{\circ}\text{C}$ has 160 times the energy density so a ship can carry a worthwhile amount	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 175-178; AQA spec 3.3.2.	
03.1	- Renewable: re-forms naturally relatively quickly so use does not necessarily reduce future availability (solar, wind, tidal, geothermal, biofuel) - Depletable: use can reduce future availability - all non-renewables (fossil fuels, uranium) plus renewables exploited unsustainably - Wood / biofuels are renewable but depletable if forests are felled faster than they regrow	3
03.2	- Other factors (technology, location, cost) restrict how much can be harnessed - Examples: coal too deep to mine; winds high above the ground / far offshore where turbines cannot be placed; waves in deep water where equipment cannot be anchored; low-intensity sunlight that cannot produce high temperatures - New technologies or higher prices may make these usable in future	3

Question	Marking guidance	Marks
03.3	<i>Any 4 from:</i> • Wind: strong reliable winds - shallow seas, open plains, uplands; low land-use conflict (not near urban areas, ecologically sensitive or scenic areas); near the grid • HEP: high reliable rainfall, large catchment, narrow valley for a small dam with a large basin, impermeable stable bedrock • Solar: high light intensity, low cloud cover • Wave: coasts with strong reliable prevailing winds and a long fetch • Tidal: large tidal range, coastal features (estuaries, channels) that focus the flow • Geothermal: hot rocks near the surface / recent volcanic activity • Biofuels: nearby farmland / forest / urban waste to avoid transporting bulky low-density fuel	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 175-177; AQA spec 3.3.2.	
04.1	• Solar PV cost fell dramatically from about 480 to 50 \$/MWh (about 90%), the steepest fall • Wind fell steadily from 200 to 45 • Gas stayed roughly constant at 60-70; solar and wind became cheaper than gas around 2015-2020 and are projected to fall further	3
04.2	• Development / research costs are still being paid for and there are no economies of scale in manufacture • They compete with established technologies whose development costs were paid long ago and which benefit from mass production and existing infrastructure • As output grows, mass production, improved designs / efficiency and experience reduce costs	3
04.3	• Most of the renewable cost is the initial equipment, paid for with loans on which interest must be paid; running costs are very low • Most of the gas project's cost is running costs (fuel) paid from income without borrowing, so it has lower interest payments	2
04.4	• Environmental damage (acid rain damage to buildings, crops, forests; health effects of air pollution; climate change) is paid for by others (health service, agriculture) not the energy user • Some impacts (climate change) cannot yet be costed; decommissioning and waste costs are also uncertain	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 179-181; AQA spec 3.3.2-3.3.3.	
05.1	• Most energy comes from non-renewable fossil fuels whose reserves are depleting; the rate of new discoveries has fallen sharply since the 1980s • Use causes unacceptable environmental damage: atmospheric pollution, climate change, oil pollution, habitat loss • Economic / social: depleting resources become more expensive, reducing future affluence; risk of an energy gap if new technologies are not developed in time	3

Question	Marking guidance	Marks
05.2	• Coal mining: habitat loss (especially open-cast), spoil heaps, subsidence from deep mines, acid drainage, dust, noise • Oil and gas extraction: spills from rigs, oil-based drilling mud polluting groundwater and sea, gas flaring releasing SO₂ / CO₂ / smoke • Pipelines causing habitat damage during construction; tanker transport risks • Fuel processing (refineries, coal washing) pollution; embodied energy of equipment and site development for all resources	4
05.3	• New technologies take many years to develop, prove and scale up, and need new infrastructure (grids, storage, vehicles) • Existing cheap fossil fuel technologies make it hard for alternatives to compete until fuel prices rise • By then depletion would reduce supply faster than replacements could be built, causing shortages	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 180-186; AQA spec 3.3.3.	
06.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Features: renewable / non-renewable, depletable, abundance, energy density, locational constraints, intermittency and predictability, need for conversions, applicability, ease of storage and transport, technological development, political and economic influences, environmental impacts • Fossil fuels: high energy density, easy storage and transport, reach high temperatures, well-developed technologies and infrastructure - but finite, polluting (acid rain, smoke, CO₂), politically concentrated (Middle East oil) • Nuclear: very high energy density, base load, few fuel deliveries - but complex, expensive, waste, public opposition • Renewables: low energy density, intermittent (wind, solar unpredictable; tidal predictable), locationally constrained, cannot be stored without conversion, need storage / grid, cannot reach fossil-fuel temperatures or replace liquid vehicle fuels yet - but renewable, low pollution, falling costs • Sustainability concepts: impacts before use (extraction, processing, equipment embodied energy) and as a consequence of use (pollution, habitat damage, depletion) • Economic and political factors: subsidies, guaranteed prices, planning, external costs not included in price, energy security • Judgement: a diverse mix matched to uses (electricity vs heat vs transport) with storage is most sustainable; transition must start before depletion	9

Question	Marking guidance	Marks
06.2	• Very high energy density fuel means only a few tens of tonnes are needed each year, compared with 10 000 t of coal per day for a coal-fired station	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 175-182; AQA spec 3.3.2-3.3.3.	